

Binary classification



Cat



Dog

$$y \in \{0, 1, 2, 3, \dots, K\}$$

Multi-class classification

More than
2
classes
=

3
Classes
=



Deer



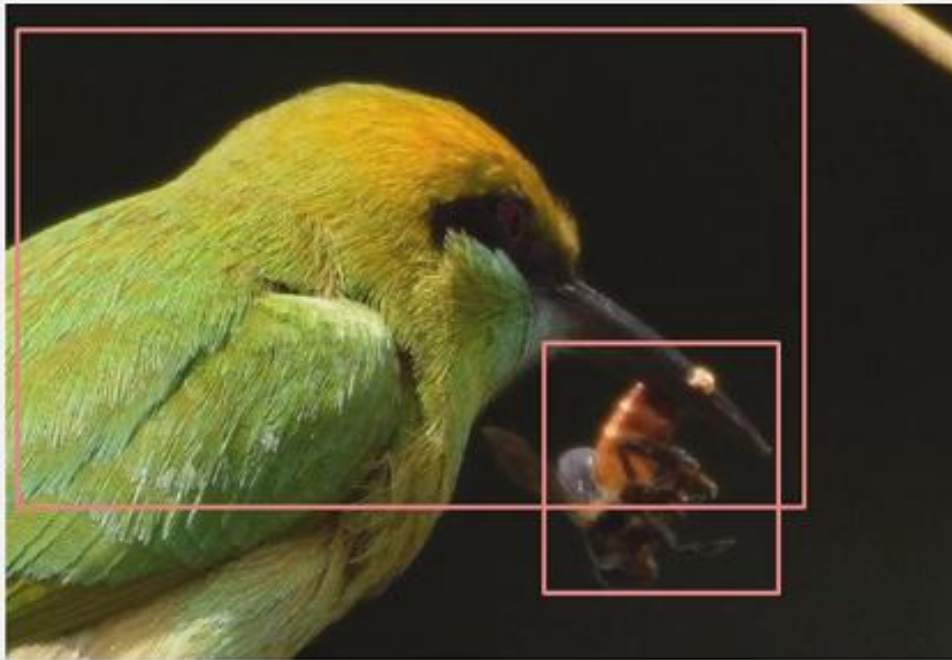
Egret



Langoor

Multi-label classification

more than one label in one sample



Classes:

Bee-eater-Bird,

Honeybee

Binary Classification Models

Logistic Regression, Support Vector Machines, Perceptron etc were designed for binary classification only.

One approach for using binary classification algorithms for multi-classification problems is to **split the multi-class classification dataset** into multiple binary classification datasets and fit a binary classification model on each. Two different examples of this approach are the **One-vs-Rest** and **One-vs-One strategies**.

One Vs Rest Strategy (OvR)

- The One-vs-Rest strategy splits a multi-class classification into one binary classification problem per class
- It involves splitting the multi-class dataset into multiple binary classification problems.
- A binary classifier is then trained on each binary classification problem and predictions are made using the model that is the most confident.